zkRep: A Privacy-Preserving Scheme for Reputation-Based Blockchain System

Chenyu Huang, Yongjun Zhao, *Member, IEEE*, Huangxun Chen, Xu Wang, Qian Zhang, *Fellow, IEEE*, Yanjiao Chen, *Senior Member, IEEE*, Huaxiong Wang, and Kwok-Yan Lam, *Senior Member, IEEE*

***Abstract*—Reputation/trust-based blockchain systems have attracted considerable research interests for better integrating Internet of Things with blockchain in terms of throughput, scalability, energy efficiency, and incentive aspects. However, most existing works only consider static adversaries. Hence, they are vulnerable to slowly adaptive attackers, who can target validators with high reputation value to severely degrade the system performance. Therefore, we introduce zkRep, a privacy-preserving scheme tailored for reputation-based blockchains. Our basic idea is to hide both the identity and reputation of the validators by periodically changing the identity and reputation commitments (i.e., aliases), which makes it much more difficult for slowly adaptive attackers to identify validators with high reputation value. To realize this idea, we utilize privacy-preserving Pedersen-commitment-based reputation updating and leader election schemes that operate on concealed reputations within an epoch. We also introduce a privacy-preserving identity update protocol that changes the identity and time-window-based cumulative reputation commitments during each epoch transition. We have implemented and evaluated zkRep on the Amazon Web Service. The experimental results and analysis show that zkRep achieves great privacy-preserving features against slowly adaptive attacks with little overhead.**

***Index Terms*—Blockchain, privacy-preserving, reputation.**

Manuscript received April 13, 2021; revised June 28, 2021; accepted August 2, 2021. Date of publication August 24, 2021; date of current version March 7, 2022. This work was supported in part by RGC under Contract CERG 16204418, Contract 16203719, Contract 16204820, and Contract R8015; in part by the Guangdong Natural Science Foundation under Grant 2017A030312008; and in part by the National Research Foundation, Singapore, under its Strategic Capability Research Centres Funding Initiative. *(Corresponding author: Qian Zhang.)*

Chenyu Huang is with Analytic Modeling Group, Ping An Technology (Shenzhen) Company Ltd., Shenzhen 518063, China (e-mail: hcyray@gmail.com).

Yongjun Zhao is with Strategic Centre for Research in Privacy-Preserving Technologies & Systems, Nanyang Technological University, Singapore (e-mail: yongjun.zhao@ntu.edu.sg).

Huangxun Chen is with the Theory Lab of 2012 Labs, Huawei, Hong Kong (e-mail: chen.huangxun@huawei.com).

Xu Wang is with the School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: wangxu298@whu.edu.cn)

Qian Zhang is with the Department of Computer Science and Engineering, Hong Kong University of Science and Technology, Hong Kong (e-mail: qianzh@cse.ust.hk).

Yanjiao Chen is with the College of Electrical Engineering, Zhejiang University, Hangzhou 310007, China (e-mail: chenyj.thu@gmail.com).

Huaxiong Wang is with the School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore (e-mail: hxwang@ntu.edu.sg).

Kwok-Yan Lam is with the School of Computer Science and Engineering, Nanyang Technological University, Singapore (e-mail: kwokyan.lam@ntu.edu.sg).

Digital Object Identifier 10.1109/JIOT.2021.3105273

I. Introduction

CURRENTLY, the integration of Internet of Things (IoT) and blockchain shows the great potential to overcome many security issues, such as point of failure, DDoS, Sybil attack and, etc. [1]–[3]. However, blockchain suffers from throughput, scalability, energy consumption, and incentive issues. Thus, the reputation/trust concept has attracted considerable interest in the combination of blockchain and IoT research community [3], [4]. The core idea of reputation is to fully consider the heterogeneity property of the IoT devices, i.e., to quantify each validators' behavior/contribution so as to provide a basis for system improvement [3]–[6]. Such behavior could be the validator's contribution of block generation [3], [5]; or the participation in the Byzantine fault tolerance (BFT) voting procedure [4]; or the trust link between each other [6]. Recent works have successfully leveraged reputation to enhance the throughput and scalability of BFT-based blockchains [4], [5] and reduce the energy costs of Proof-of-Work (PoW) systems [3], [6]. Moreover, introducing reputation makes it smoother to integrate with incentive schemes [4], [6].

However, existing reputation-enhanced blockchains [3]–[6] only consider the static adversary. Thus, they are particularly vulnerable to slowly adaptive attackers, who can first observe the system, and then choose a validator with a high reputation as the victim. This results in two serious problems. First, reputation fails to be a reliable indicator of honest and high capability validator. Second, the system performance may degrade dramatically due to the corruption of high reputation validators. Slowly adaptive attacks have been studied in other blockchain systems [7]–[10], but have never been investigated in reputation-enhanced ones. The great benefits of reputation shown in previous efforts [3]–[6] motivate us to fill this gap.

Based on the above analysis, we find the key to mitigating slowly adaptive attacks is to conceal validators' reputation values (i.e., treat reputation as privacy). In this way, the attacker cannot identify high-value targets (i.e., validators with a high reputation) to degrade the system. In other words, it is desired to develop an effective privacy-preserving scheme for reputation-enhanced blockchains. To our knowledge, none of the previous efforts on privacy-preserving techniques works harmoniously with reputation-enhanced blockchains. Mixing services [11], [12] and ring signature [13], [14] cannot support reputation-enhanced leader election [4], [5]. Zero-knowledge proof (ZKP)-based methods [15]–[17] focus on

2327-4662 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

clients' privacy rather than validators' privacy. Works [18]–[20] are designed to protect the privacy of the stake in PoS-based blockchains. The values of stake are concealed in the transactions, i.e., unobservable to the public. However, reputation values are computed based on valuators' behaviors that are transparent to the public. Such behaviors could be the times of successfully mining the block [3], [5], voting behavior [4], the trust reported by a validator's neighbors [6]. These behaviors can be viewed as "plaintext," (i.e., transparent) and cannot be hidden due to the need for reputation calculation/consensus, which makes it infeasible to adopt techniques in [18]–[20].

To fill this gap, we propose a privacy-preserving scheme, zkRep, tailored for reputation-enhanced blockchains. The key idea to defend against slowly adaptive attacks is to hide both the identity and reputation of the validators by changing their identity and reputation commitments (i.e., aliases) in an epoch-based manner. This dramatically increases the level of difficulty for a slowly adaptive attacker to identify the validators with high reputation values. Although the idea looks straightforward, we need to tackle two major challenges to fully realize the system: 1) how to update reputation values and elect leaders with concealed reputations (i.e., reputation commitments) within an epoch and 2) how to securely and anonymously change the identity/reputation commitments during epoch transition to successfully defend against slowly adaptive attackers.

We propose two schemes to address these challenges. First, we propose the Pedersen-commitment-based reputation updating and leader election schemes. The reputation is hidden in the Pedersen commitment, which still supports update operations, including adding and decaying. In addition, we propose two ZKP-based leader election schemes to elect a leader based on the concealed reputation, one for reputation-enhanced BFT-based blockchains [4], [5] and the other for reputation-enhanced PoW-based blockchains [3], [6]. Second, we propose a privacy-preserving identity update (PIU) protocol that changes validators' IDs and reputation commitments via ZKP during each epoch transition. As a result, slowly adaptive attackers cannot link the old and new IDs/reputations, because our schemes make it hard for them to learn reputation through continuous observation. In addition, we use the time-window-based cumulative reputation to balance the effects of the historic and recent reputation in the ID update protocol.

We tested our proof-of-concept implementation on Amazon Web service (AWS) with 100 instances. The results show our scheme achieves great privacy-preserving features with little overhead to the original reputation-enhanced blockchains. Extensive analysis and evaluations demonstrate that zkRep can further enhance the reputation-based blockchain [3]–[6] to defend against slowly adaptive attackers and preserve the benefits of reputation to the system performance.

The contributions of this article are summarized as follows.

1) We take the first step to designing a privacy-preserving scheme, zkRep, tailored for reputation-based blockchains.
2) We propose a privacy-preserving Pedersen-commitment-based reputation updating protocol and two ZKP-based leader election schemes to operate on concealed reputations, for reputation-enhanced BFT-based and PoW-based blockchains, respectively.
3) We propose a PIU protocol to securely and anonymously change identity/reputation commitments to defend against slowly adaptive attackers.
4) Based on extensive evaluation on AWS, we provide experimental results and detailed analysis to show that zkRep can defend against slowly adaptive attacks with little overhead.

II. Related Work

A. Reputation-Based Blockchains

Current blockchains can hardly meet many requirements in IoT scenarios that have scalability, high throughput, and energy constraints. Thus, many reputation-based blockchains have been developed. The core idea of such a paradigm is to measure the behavior of the IoT devices, and carefully design the consensus protocol that fully utilizes such heterogeneity. RepuCoin [5] achieves consensus based on a combination of PoW and reputation weighted BFT. RepChain [4] adopts a reputation-based blockchain with sharding. The validators are divided into several balanced shards, and the leader is selected based on its reputation to improve the throughput and scalability of the blockchain. B-IoT [3] and PoT [6] both utilize a proof-of-trust/credit-based PoW. The difficulty of PoW is determined by the reputation of the nodes. B-IoT adopts the directed acyclic graph (DAG) to measure the reputation while PoT gets it from a trust graph that is reported by nodes in the network. However, none of these blockchains use the privacy-preserving technology and thus can be defeated by a slowly adaptive attacker. One may regard the reputation-based blockchain as similar to the PoS-based blockchain. However, there are two key differences: 1) reputation mainly comes from the transparency behavior, i.e., the times the validator successfully mining the block or the voting times in the BFT consensus, while PoS is mainly from the stake and 2) PoS usually needs the underlying crypto-currency while reputation-based blockchain is more flexible without such requirement [6].

B. Privacy-Preserving Blockchains

After the release of Bitcoin [21], various blockchains have been proposed. However, the number of transactions and the links between them are public and can be tracked. Thus, privacy-preserving blockchain systems try to tackle the problem. P2P mixing service [11], [12] mixes multiple input and output addresses of a transaction so that an adversary cannot link the input and output. However, the adversary can corrupt the service by inserting fake data during the mixing. Although the protocol still remains secure by restarting, the cost of communication and computation will increase a lot. Ring signature [13], [14] is also an attractive approach. It can hide the actual spender among multiple valid participants. However, it is difficult to build the privacy-preserving reputation-based lottery procedure from Ring signatures. Algorand [22] leverages verifiable random functions

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

to privately agree on the next block. It avoids targeted attacks at chosen participants using participant replacement at every step. However, it does not anonymize the stake of the owner.

ZKP is the most widely used method to anonymize users. Zerocoin [15] and Zerocash [16] use ZKP to hide the information of transactions. Along this line, zkLedger [17] proposes an auditable blockchain based on standard noninteractive ZKP with the help of audit tokens. However, these works only consider the privacy of the users but not the validators. The most relevant works are PoS-based privacy-preserving blockchains. Ouroboros Crypsinous [18] proposes a privacy-preserving PoS protocol relying on zk-SNARKs and key-private forward-secure encryption. Its protocol determines the leader based on individual split coins which can be revealed. Ganesh *et al.* [19] proposed a private leader election protocol based on an anonymous verifiable random function. Baldimtsi *et al.* [20] proposed a general anonymous selection for PoS and instantiate it with Algorand [22]. However, they cannot be used directly in reputation-based blockchains due to the transparency property of reputation.

III. Overview

A. Reputation-Based Blockchains

Each validator of a reputation-based blockchain has an attribute called "reputation." Different reputation-based blockchains may determine the reputations of their participants in different ways. There are two main approaches in the literature. 1) Reputation values are determined by participants' contribution to the block generation. B-IoT [3] and RepuCoin [5] reward the credit to the block miners. RepChain [4] calculates the reputation based on the validators' voting in the BFT consensus. 2) Reputation is quantified by the links between validators. E.g., in PoT [6], reputation is calculated via the trust links announced by a validator. Such a link represents its trusted neighbors in a P2P situation. We focus on approach 1) for illustration, but zkRep is compatible with approach 2) as well. See Section VI-C for discussion.

Recording reputation value in a blockchain indicates that consensus on this reputation has been reached. There are two main kinds of consensus on reputation: 1) credit-based PoW [3], [6] in which validators mine new blocks with different difficulty levels according to their reputations, so that the system produces new blocks with less latency and energy and 2) reputation-based BFT [4], [5] whose consensuses are based on reputation-weighted voting [5] or reputation-based leader election [4], so as to enhance the throughput and security compared with nonreputation-based blockchains. Table I compares existing reputation-based blockchains.

TABLE I
Comparison Between Reputation-Based Blockchains

System	Consensus	Communication Model	Permissioned?
B-IoT	Credit-based PoW	Asynchrony	Yes
PoT	Credit-based PoW	Asynchrony	No
RepuCoin	PoW + Reputation-based BFT	Partial Synchrony	No
RepChain	Reputation-based BFT	Synchrony	Yes

B. System Overview

zkRep is an epoch-based blockchain and Fig. 1 shows the overview, including three parts.

1) Privacy-Preserving Reputation: At the beginning of an epoch, the validators should submit their ID and reputation commitments. zkRep leverages Pedersen commitments to preserve the privacy of IDs and reputations. In addition, it guarantees the consensus on the update operations (add/decay) on committed reputations within an epoch can be reached.

2) Privacy-Preserving Leader Election: Within an epoch, an elected leader adds a block to record updated reputation commitments. Update (add/decay) and record operations repeat alternately to form blockchain before an epoch ends. In zkRep, the probability that a validator is elected to be a leader is proportional to its reputation. We avoid leaking the information about the elected leader via ZKP and propose two privacy-preserving leader election (PLE) schemes, one for reputation-based BFT and the other for credit-based PoW.

3) Privacy-Preserving Identity Update Protocol: To defend against slowly adaptive attackers, we propose a PIU protocol via ZKP. It anonymously renews the commitments of each validator's identity and reputation during epoch transitions. Also, the protocol balances the effects of historic and recent reputation via time-window-based cumulative reputation.

IV. System Preliminaries

A. System Model

zkRep is built on the reputation-based blockchain that is processed in a fixed time called *epochs*. We assume a permissionless model that demands Sybil-resistant identities, similar to those in existing epoch-based blockchains [7], [8]. In zkRep, any node can join the system at epoch e by submitting a PoW puzzle solution for eligible identity generation. All these identities can be recorded in the first block of each epoch to achieve consensus, i.e., the newly joined validator should work from epoch $e+1$ and the number of validators within an epoch is static. Without loss of generality, we assume there are n validators $V^{n_e} = \{v_1^e, v_2^e, \ldots, v_n^e\}$ in one epoch, who process the transactions and generate new blocks. The reputation of validator i at block k is $R_{i,k}^e$. And the newly joined validator will have a predefined initial reputation R_{init}.

In this way, we separate the consensus to generate identities and the consensus to process transactions, i.e., the enabling flexible intraepoch scheme choosing between reputation-based BFT and credit-based PoW under the permissionless setting. All involved variables in zkRep are listed in Table II.

B. Network Model

In zkRep, the network model for PLE depends on the intraepoch consensus. Credit-based PoW works under an asynchronous model as in B-IoT and PoT in Table I. For BFT, we assume a synchronous network model which is the same as that in RepChain. The network model within epoch and during epoch transition can be different which is common in

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

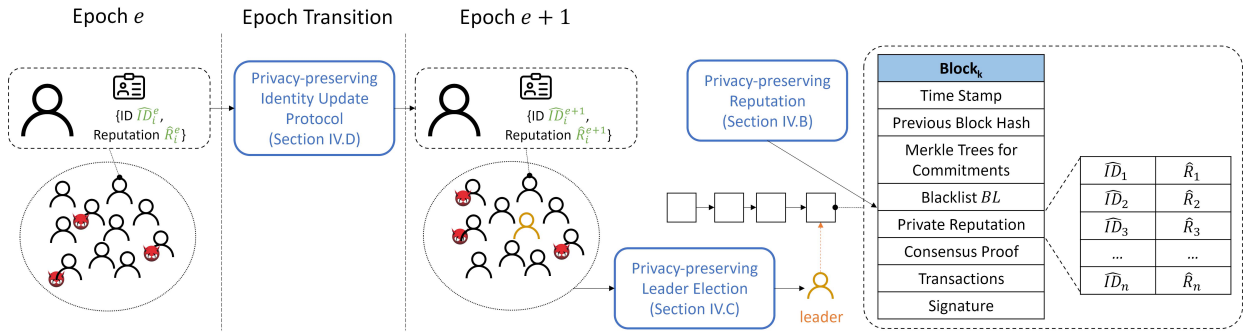

Fig. 1. System overview of zkRep and the block structure.

TABLE II
SUMMARY OF VARIABLES IN ZKREP

n, f	number of validators and Byzantine adversaries
e	epoch index
$\widehat{ID}_i^e$	ID of validator i in epoch e
$R_{i,k}^e$	reputation of validator i in epoch e at block k
$\Delta R_{i,k}^e$	reputation earned by validator i between block $k-1$ and k in epoch e
ΔR_i^e	reputation earned by validator i in epoch e
TR_k^e	total reputation of all the validators in epoch e at block k
$\widehat{R}_{i,k}^e$, $\Delta\widehat{R}_{i,k}^e$	commitment of $R_{i,k}^e$ and $\Delta R_{i,k}^e$ respectively
$\widehat{RD}_{i,k}^e$	commitment of $R_{i,k}^e$ after applying the decaying function $\overline{\mathsf{N}(t)}$
TD_x	trust division x
w	size of the sliding time window
rs_i	root secret of validator i
r_i^e, s_i^e	commitment random number of validator i
rt_{ID}^e, rt_R^e	root hash of the Merkle-tree for ID/reputation commitments in epoch e
$\pi_{\mathtt{PPOW}}$, $\pi_{\mathtt{PLE}}$ $\pi_{\mathtt{PIU}}$	proofs for NP statements PPOW, PLE and PIU

epoch-based blockchains. Thus, for PIU protocol during epoch transition, we assume synchronous channels since the message delay can be in the scale of the minutes level. In addition, we assume each validator has a pair of public/private keys in each epoch, and all messages sent in the network are authenticated with the sender's private key.

C. Threat Model

The adversaries considered in zkRep are the intersection of probabilistic polynomial-time (PPT) Byzantine adversaries and slowly adaptive Byzantine adversaries. Given n validators in an epoch, a PPT Byzantine adversary corrupts f validators before the epoch, and the corrupted nodes can send malformed messages, remain silent, and collude with each other. For two intraepoch consensuses, reputation-based BFT and credit-based PoW, zkRep sets different f in the threat model: $f < n/3$ for the former; and the total reputation of highest f validators should be less than $1/2$ total reputation for the latter.

In terms of slowly adaptive behaviors, zkRep adopts the same setting as in committee-based blockchains [7]–[10] can adaptively choose victims to compromise on the order of epochs. Specifically, the attacker can determine and compromise the victims during epoch transition. Then, it can observe all the validators' behavior within the epoch but cannot change the targets. The corrupted nodes can behave arbitrarily and collude with each other, e.g., they may send malformed messages or remain silent. The consensus can be stuck when the adversary is wrongly elected as a leader, leading to the decrease of the whole system's throughput. In reputation-based blockchains, the strategy of the attacker is to purposely choose the high-reputation validators as victims.

D. Cryptographic Building Blocks

1) Pedersen Commitment and Pedersen Hash: We use both to hide the reputations and IDs of validators. These functions are based on a twisted Edwards curve named JubJub and thus can be calculated efficiently in zk-SNARKs. We denote $\mathsf{COMM}(\cdot,\cdot)$ as the Pedersen commitment and $\mathsf{H}(\cdot)$ as the Pedersen hash. The Pedersen commitment has the following properties: 1) *additively homomorphic:* given $cm_1 = \mathsf{COMM}(m_1, r_1)$ and $cm_2 = \mathsf{COMM}(m_2, r_2)$, one can compute the commitment of $m_1 + m_2$ via $cm_1 + cm_2$; 2) *perfectly hiding:* for any commitment, every m' is equally likely to be the committed secret value; and 3) *computationally binding:* It is infeasible for a PPT adversary to output $(m, r), (m', r')$ such that $\mathsf{COMM}(m, r) = \mathsf{COMM}(m', r')$ and $m' \neq m$. The Pedersen hash is a collision-resistant hash function, i.e., it is infeasible to find $x \neq y$ such that $\mathsf{H}(x) = \mathsf{H}(y)$.

2) Zero-Knowledge Succinct Noninteractive Argument of Knowledge: To assert ownership and leadership, a validator should convince others without revealing its own reputation. In the literature, many ZKP systems can be used to construct such an assertion. In zkRep, we adopt zk-SNARKs [23] to show the effectiveness. Specifically, let C be a nondeterministic decision circuit for an NP-language $\mathsf{L_C}$. zk-SNARK has three functions.

1) *KeyGen*$(1^\lambda, C) \rightarrow (pk, vk)$*:* Given an input circuit C and a security parameter λ, it generates a proving key pk and a verification key vk via a one-time trusted setup.
2) *Prove*$(pk, x, a) \rightarrow \pi$*:* On input pk and a witness a for the statement $x \in \mathsf{L_C}$, outputs a proof π.
3) *Verify*$(vk, x, \pi) \rightarrow b$*:* The proof π can be verified by any other validators with the vk and x. The output $b = 1$ indicates that the statement $x \in \mathsf{L_C}$.

zk-SNARK has four security properties: 1) completeness; 2) succinctness; 3) soundness; and 4) perfect zero-knowledge [23].

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

V. Design

A. Privacy-Preserving Reputation

We first illustrate the block structure of the reputation blockchain and the structures of ID and reputation commitments. Then, we introduce the realization of a decaying operation on committed reputation.

1) Privacy-Preserving Reputation Block: A reputation-based blockchain records the validators' reputation values on the chain. The general structure of the block is shown in Fig. 1. The field "Consensus Proof" is the proof to validate the block, e.g., it could be the "nonce" field in the PoW-based blockchain [6] or the collective signature in BFT protocol [5]. The field "Private Reputation" denotes the validators' reputations that have achieved consensus. It contains two fields: 1) ID commitment and 2) reputation commitment of validator i. To hide both ID_i and R_i, we commit them as $\widehat{ID}_i^e$ and $\widehat{R}_{i,k}^e$ at epoch e where k is the index of the block. $\widehat{ID}_i^e$ remains the same for validator i within epoch e, while the $\widehat{R}_{i,k}^e$ changes based on the behavior of validator i. Specifically, the validator i calculates reputation commitment $\widehat{R}_{i,0}^e = \mathsf{COMM}(R_{i,0}^e + q, r_i^e)$ at the beginning of epoch e. Then it calculates the ID commitments $\widehat{ID} = \mathsf{COMM}(\mathsf{H}(R_{i,0}^e || rs_i || pk_i^e || e), s_i^e)$, where pk_i^e is the public key and rs_i is the root secret of validator i (refer to Section V-C for details). The root secret rs_i could be a random number chosen by the validator i. The purpose of using $R_{i,0}^e$ in ID commitments' calculations is to link the reputation and ID in epoch e. Then the validator submits both commitments, $\widehat{ID}_i^e$ and $\widehat{R}_{i,k}^e$. It is worth mentioning that the reputation could be decreased due to the penalty on the malicious behavior. Thus, the constant q is used to avoid negative reputations. The "Merkle Trees for Commitments" and "Blacklist" fields will be described in Section V-C. The former one is used to store the reputation and identity commitments and the latter one is used to store the exposed root secret. The validators may be compromised by the attackers and their root secret could be abused. In this case, the victim should not use the root secret and reveal it once it is recovered. All other validators will help to record the exposed secret root in the "Blacklist" field.

In this article, we do not focus on specific methods of behavior monitoring and reputation calculation which have been well investigated in the literature. We only focus on general reputation updating, in which all the validators need to add the reputation $\Delta R_{i,k}^e$ to $\widehat{R}_{i,k-1}^e$. Since all the validators should reach consensus on the new reputation commitment of i and write it on the block, they will add $\mathsf{COMM}(\Delta R_{i,k}^e, 0)$ to $\widehat{R}_{i,k-1}^e$. Due to the homomorphic property of the Pedersen commitment, $\widehat{R}_{i,k}^e = \widehat{R}_{i,k-1}^e + \mathsf{COMM}(\Delta R_{i,k}^e, 0) = \mathsf{COMM}(R_{i,k}^e + q, r_i^e)$. As a result, all the validators will have the same $\widehat{R}_{i,k}^e$ without leaking $R_{i,0}^e$ as well as $R_{i,k}^e$.

2) Reputation Decaying: Reputation decaying is an important feature in reputation-based blockchains to prevent high reputation validators from monopolizing the system. In this work, we adopt a similar decaying function as PoT [6]. Specifically, a decaying penalty $N(t) = e^{\eta t}$, where t denotes time and η denotes a decay constant, is applied to a validator's reputation when it becomes the leader. Recall that the message space includes only positive integers, so we leverage the fixed-point representation [24], [25] as $\overline{\mathsf{N}(t)} = \lfloor 2^p \cdot (1 - \mathsf{N}(t)) \rfloor$, where the term 2^p is a fixed resolution.

NP Statement 1 PPOW

Given the ID commitment $\widehat{ID}_i^e$, reputation commitment $\widehat{RD}_{i,k}^e$, epoch e, the block content $B_k'^e$ without the consensus proof field, and the time t in decaying function, we can know the nonce that makes block B_k^e valid such that:

- $\widehat{RD}_{i,k}^e$ and $\widehat{ID}_i^e$ are well-formed: $\widehat{RD}_{i,k}^e = \mathsf{COMM}((R_{i,k}^e + q) \cdot \overline{\mathsf{N}(t)}, r_i^e \cdot \overline{\mathsf{N}(t)})$, $\widehat{ID} = \mathsf{COMM}(\mathsf{H}(R_{i,0}^e || pk_i^e || e), s_i^e)$.
- The trust division TD_x of i is correct: $R_{i,k}^e \cdot \overline{\mathsf{N}(t)} \in TD_x$
- The block is valid by a range proof: $\mathsf{H}(B_k'^e || nonce) \leq \mathsf{diff}(TD_x)$.

For every block k, zkRep applies the decaying function to the recent leaders, e.g., when the validator i is the leader of the block k' ($k - \alpha \leq k' < k$, where α is the number of blocks for which the validators are subjected to the decaying procedure), its reputation at block k will be $R_{i,k}^e \cdot \overline{\mathsf{N}(k-k')}$. Thus, the committed reputation is $\widehat{RD}_{i,k}^e = (\widehat{R}_{i,k}^e)^{\overline{\mathsf{N}(k-k')}} = \mathsf{COMM}((R_{i,k}^e + q) \cdot \overline{\mathsf{N}(k-k')}, r_i^e \cdot \overline{\mathsf{N}(k-k')})$. For a nonleader validator, its reputation is $(R_{i,k}^e + q) \cdot \overline{\mathsf{N}(0)} = (R_{i,k}^e + q) \cdot 2^p$. For simplicity of calculating the total reputation over all the validators, the decaying function is only used within the epoch.

B. Privacy-Preserving Leader Election

In reputation-based blockchains, the probability of a validator being elected as the leader in an epoch is proportional to its reputation. zkRep proposes two PLEs via zk-SNARKs for two main consensuses, credit-based PoW, and reputation-based BFT, respectively.

1) Privacy-Preserving Credit-Based PoW: Both B-IoT and PoT use the credit-based PoW. Specifically, the reputations are grouped into different trust divisions TD with different reputation ranges. For each TD, the difficulty level $\mathsf{diff}(TD_x)$ is different. Thus, a block B_k^e proposed by a validator in trust group TD_x is valid iff $\mathsf{H}(B_k^e) \leq \mathsf{diff}(TD_x) \wedge R_{i,k}^e \in TD_x$.

zkRep mines the nonce and hides the reputation of privacy-preserving credit-based PoW based on zk-SNARK. zkRep first deploys the $pk_{\mathtt{PPOW}}$ and $vk_{\mathtt{PPOW}}$ generated by a one-time setup $\mathsf{KeyGen}(1^\lambda, C_{\mathtt{PPOW}})$. Then within epoch e, validator i will mine a *nonce* of privacy-preserving credit-based PoW as Algorithm 1 shows. Specifically, a validator i first mines a block by iterating the *nonce*. Once it finds a valid block, it produces a zk-SNARK proof of credit-based PoW, $\pi_{\mathtt{PPOW}}$. The NP statement of $\pi_{\mathtt{PPOW}}$ called PPOW is shown in NP Statement 1.

Then, the validator i will fill the *nonce* and $\pi_{\mathtt{PPOW}}$ into the consensus proof field of block k and broadcast the block. Other validators can verify the consensus proof by running $b = \mathsf{Verify}(vk_{\mathtt{PPOW}}, \vec{x}, \pi_{\mathtt{PPOW}})$. The block will be valid iff $b = 1$.

2) Privacy-Preserving Reputation-Based BFT Leader Election: The BFT-based leader election selects the leader via a reputation weighted lottery [4]. Specifically, the validators get a random number $rn_{i,k}^e$ for validator i from a random number generator with the same seed. Such seed could be

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 PPOW for Validator i

Input:
The epoch e, the reputation $R^e_{i,k}$, the random numbers r^e_i and s^e_i, block content without consensus field B'^e_k, t in decaying function and the ID commitment $\widehat{ID}^e_i$.

Output:
The proof $\pi_{\mathtt{PPOW}}$.

1: $nonce = 0$, $isValid = \mathsf{false}$.
2: Determine the TD_x based on the $R^e_{i,k}$.
3: **while** $!isValid$ **or** $!TimeOut$ **do**
4: **if** $\mathsf{H}(B'^e_i||nonce) \leq \mathsf{diff}(TD_x)$ **then**
5: $isValid = \mathsf{true}$.
6: **else**
7: $nonce = nonce + 1$.
8: **end if**
9: **end while**
10: **if** $isValid$ **then**
11: Set $\vec{a} = (R^e_{i,k}, r^e_i, s^e_i, nonce)$
12: Set $\vec{x} = (\widehat{ID}^e_i, \widehat{RD}^e_{i,k}, B'^e_i, e, t)$
13: Compute $\pi_{\mathtt{PPOW}} = \mathsf{Prove}(pk_{\mathtt{PPOW}}, \vec{x}, \vec{a})$
14: **end if**

generated by the secure distributed bias-resistant randomness generation protocol [8]. Then, the validators calculate the float number $p_i = R^e_{i,k}/rn^e_{i,k}$ and the validator with the smallest p_i wins the lottery and becomes the leader.

zkRep hides the reputation and elects the leader by a privacy-preserving lottery based on zk-SNARK. zkRep first deploys the $pk_{\mathtt{PLE}}$ and $vk_{\mathtt{PLE}}$ generated by a one-time setup $\mathsf{KeyGen}(1^\lambda, C_{\mathtt{PLE}})$. Within epoch e before block k, each validator will get the random number $rn^e_{i,k}$, the range of which has the maximum length of l_{rn}. Then the validators will run the privacy-preserving BFT leader election as Algorithm 2 shows. To avoid a low reputation validator from winning, we add a constraint that the $R^e_{i,k}$ of the leader must be larger than the mean reputation. The requirement is captured by the range proof $R^e_{i,k} \cdot n \geq TR^e_k$. The validator i wins the lottery if $rn^e_{i,k}/2^{l_{rn}} \leq R^e_{i,k}/TR^e_k$ (line 1). TR^e_k is the total reputation of all the validators. Recalling that the decaying the function is only used within the epoch and the validator's ID will not change within the epoch, the total reputation can be easily calculated by summing the ΔR^e and adding the decaying function on the historical leader. However, the possibility that multiple validators win the lotteries exists. Thus, each validator should calculate a random number $rnc^e_{i,k}$ (line 2) if it wins the lottery. Then validator i generates a zk-SNARK proof of the privacy-preserving BFT leader election, $\pi_{\mathtt{PLE}}$. The NP statement of $\pi_{\mathtt{PLE}}$ called PLE is shown in NP Statement 2.

After the proof is generated, the validator i will fill $\pi_{\mathtt{PLE}}$ and $rnc^e_{i,k}$ into the consensus proof field of block and broadcast it. Other validators can validate the block by running $b = \mathsf{Verify}(vk_{\mathtt{PLE}}, \vec{x}, \pi_{\mathtt{PLE}})$. The block is valid iff $b = 1$. Then all the validators will run the BFT consensus to finally add the valid block into the blockchain. If multiple valid blocks are received, the validators will select the block with the minimum $rnc^e_{i,k}$. Note that if no one wins the lottery, zkRep will call

NP Statement 2 PLE

Given the $\widehat{ID}^e_i$, $\widehat{RD}^e_{i,k}$, epoch e, t in the decaying function, the random number $rn^e_{i,k}$, we know that validator i wins the lottery with a random number $rnc^e_{i,k}$ such that:

- $\widehat{RD}^e_{i,k}$, $\widehat{ID}^e_i$ and random number $rnc^e_{i,k}$ are well-formed: $\widehat{RD}^e_{i,k}$ and $\widehat{ID}^e_i$ are the same as that in PPOW, $rnc^e_{i,k} = \mathsf{H}(rn^e_{i,k}||\widehat{R}^e_{i,k}||B'^e_{i,k})$.
- Range proof for leader candidate: $R^e_{i,k} \cdot n \geq TR^e_k$.
- Range proof for lottery: $rn^e_{i,k} \cdot TR^e_k \leq 2^{l_{rn}} \cdot R^e_{i,k} \cdot \overline{\mathsf{N}(t)}$.

Algorithm 2 Privacy-Preserving BFT Leader Election for Validator i

Input:
The epoch e, the reputation $R^e_{i,k}$, the random number rn^e_i, the length l_{rn}, block content without consensus field B'^e_k, t in decaying function and the ID commitment $\widehat{ID}^e_i$.

Output:
The proof $\pi_{\mathtt{PLE}}$, and a random number $rnc^e_{i,k}$.

1: **if** $rn^e_{i,k} \cdot n \geq TR^e_k$ **and** $rn^e_{i,k} \cdot TR^e_k \leq 2^{l_{rn}} \cdot R^e_{i,k} \cdot \overline{\mathsf{N}(t)}$ **then**
2: Compute $rnc^e_{i,k} = \mathsf{H}(rn^e_{i,k}||\widehat{RD}^e_{i,k}||B'^e_{i,k})$
3: Set $\vec{a} = (R^e_{i,k}, r^e_i, s^e_i)$
4: Set $\vec{x} = (\widehat{ID}^e_i, \widehat{RD}^e_{i,k}, B'^e_i, e, TR^e_k, t, rnc^e_{i,k})$
5: Compute $\pi_{\mathtt{PLE}} = \mathsf{Prove}(pk_{\mathtt{PLE}}, \vec{x}, \vec{a})$
6: **end if**

the BFT consensus to re-elect the leader. Specifically, after the timeout, the validators will generate an empty block that contains a null Merkle root and current block number k. Then, the validators will use BFT consensus to agree on this block and enter the next iteration to generate block $k+1$ and re-elect the leader for this block.

C. Privacy-Preserving Identity Update Protocol

zkRep can elect the leader without leaking the information of validators. However, a slowly adaptive attacker can still break the system by periodically changing the victims and inferring the relationship between the reputations and IDs. In addition, the consensus is based on the plaintext $\Delta R^e_{i,k}$. Thus, the attacker can gradually obtain all reputation scores. Thus, we need an appropriate PIU protocol that periodically changes the ID and reputation commitments. Such a protocol should have the following features: 1) the ID and reputation should be updated without exposing validators and 2) the reputation should be calculated based on limited recent duration to prevent monopoly [3], [4]. The protocol security depends on the epoch length, i.e., with a longer epoch length, it is more likely that the attacker can infer the high reputation validators from the frequency of block generation. In Section VI-B4, we give a detailed analysis and propose a scheme to set the appropriate epoch length.

1) Basic Privacy-Preserving Identity Update: To realize feature 1) we propose a basic PIU protocol as in Fig. 2. Specifically, zkRep first deploys the $pk_{\mathtt{PIU}}$ and $vk_{\mathtt{PIU}}$ generated by a one-time setup $\mathsf{KeyGen}(1^\lambda, C_{\mathtt{PIU}})$. At the beginning of epoch $e-1$, the validators will generate two Merkle-trees

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

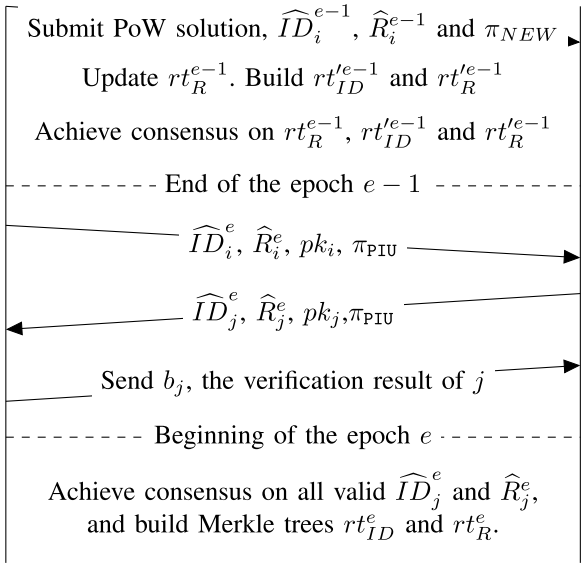

Fig. 2. Basic PIU protocol for a new joined validator *i*.

with roots rt_{ID}^{e-1} and rt_R^{e-1} in the first block. The former is for the ID commitment, and the latter is for the reputation commitment. At the end of the epoch $e-1$, all the validators will update rt_R^{e-1} based on the final reputation R_i^{e-1} of validator *i*. Some new comers may join the system. Thus, the validators will also build up two Merkle-trees with roots rt'^{e-1}_{ID} and rt'^{e-1}_R for the newcomers. The legality of these Merkle-trees will be guaranteed by running another round of consensus algorithm. And they will be stored in the last block of epoch $e-1$.

A newcomer *i* can join the system by submitting: 1) their ID and reputation commitments; 2) a zk-SNARK proof π_{NEW}; and 3) a solution of PoW puzzle within the epoch $e-1$. Specifically, π_{NEW} proves the NP statement that the ID commitment is well-formed and the reputation commitment is $\mathsf{COMM}(R_{\text{init}}+q, r_i^{e-1})$ where R_{init} is the initial default reputation score. The solution of the PoW puzzle is used to prevent Sybil attack, it can be generated by finding a *nonce* so that $H(\hat{ID}_i^e||e-1||nonce||seed^{e-1}) < D$ where D is the predefined difficulty and the $seed^{e-1}$ is the global random seed. The global random seed can be generated during epoch transition before epoch $e-1$ by secure distributed randomness generation protocol such as RandHound [26].

Before the beginning of epoch *e*, all validators should produce a proof, $\pi_{\mathtt{PIU}}$, of a new ID commitment $\widehat{ID}_i^e$ and reputation commitment $\widehat{R}_{i,k}^e$. The NP statement of $\pi_{\mathtt{PIU}}$ called PIU is shown in NP Statement 3. After validator *i* broadcasts $\widehat{ID}_i^e$, $\widehat{R}_{i,0}^e$, and $\pi_{\mathtt{PIU}}$. Other validators will verify it by running $b = \mathsf{Verify}(vk_{\mathtt{PIU}}, \vec{x}, \pi_{\mathtt{PIU}})$, where $\vec{x} = (\widehat{ID}_i^e, \widehat{R}_{i,0}^e, e, rt_{ID}^{e-1}, rt_R^{e-1}, rt'^{e-1}_{ID}, rt'^{e-1}_R)$. The proof is valid iff $b = 1$. Then, the validators will broadcast *b* and achieve consensus for all the valid ID and reputation commitments.

At the beginning of epoch *e*, all the validators will generate rt_{ID}^e and rt_R^e for all valid $\widehat{ID}_j^e$ and $\widehat{R}_j^e$, respectively. Since the $\widehat{ID}_i^e$ have the same rs_i with $\widehat{ID}_i^{e-1}$, such proof can prove this validator is either valid in the last epoch or a valid newcomer.

NP Statement 3 PIU

Given the $\widehat{ID}_i^e$, $\widehat{R}_{i,0}^e$, epoch *e* and the Merkle-tree roots rt_{ID}^{e-1}, rt_R^{e-1}, rt'^{e-1}_{ID} and rt'^{e-1}_R, we know that the validator *i* is valid with new commitments in epoch *e* such that:

- $\widehat{ID}_i^e$ and $\widehat{R}_{i,0}^e$ are well-formed.
- $\widehat{ID}_i^{e-1}$ has the same rs_i as $\widehat{ID}_i^e$.
- Root secret is not in the blacklist: $rs_i \notin BL$.
- $\widehat{RD}_{i,k}^{e-1}$ has the same reputation as $\widehat{R}_{i,0}^e$: $R_{i,0}^e = R_i^{e-1}$
- The $\widehat{ID}_i^{e-1}$ and $\widehat{RD}_{i,k}^{e-1}$ either appear as the leaf of a Merkle-tree with root rt_{ID}^{e-1} and rt_R^{e-1} respectively, or appear as the leaf in rt'^{e-1}_{ID} and rt'^{e-1}_R respectively.

Note that the rs_i is the root secret for generating the ID and reputation commitments. Once it has been stolen, the attacker can inherit the reputation from the victim. To prevent this, the victim *i* can reveal the rs_i by sending it with an identity commitment $\hat{ID}_i^{e'}$ generated by it. The victim should also sends related parameters($e', R_{i,0}^{e'}, pk_i^{e'}, r_i^{e'}, s_i^{e'}$) so that the leaked root secret can be easily verified if: 1) the $\hat{ID}_i^{e'}$ is generated by it and 2) $\hat{ID}_i^{e'}$ appears in the $rt_I^{e'}D$. The other validators can store rs_i in the "Blacklist" *BL* field of the block. A validator needs to prove its commitments are not generated from the blacklist *BL* in PIU.

2) Time-Window-Based Reputation: With the basic scheme, the reputation considers the whole history of the validator which may cause the monopoly. To realize feature 2), zkRep aggregates the reputation within a limited time window that covers *w* recent epochs. At the beginning of epoch $e-1$, the validators will generate Merkle-trees that contain the reputation commitments of recent *w* epochs. Specifically, there are $w-1$ Merkle-trees with roots of $rt_{\Delta R}^{e-j}(1 < j \leq w)$ which contain the reputation commitment $\Delta\widehat{R}_{i,j}^{e-1} = \mathsf{COMM}(\Delta R_i^{e-j} + q, r_i^{e-1})$. The validator *i*'s $\Delta\widehat{R}_{i,j}^{e-1}$ should have the same r_i^{e-1} for total reputation commitments calculation. The Merkle-trees with root rt_{ID}^{e-1} for ID commitments will also be built. At the end of the epoch, the validators will generate the $rt_{\Delta R}^{e-1}$ as well as rt'^{e-1}_R and rt'^{e-1}_{ID}. The proof is similar to $\pi_{\mathtt{PIU}}$. But it needs to prove $\Delta\widehat{R}_{i,j}^{e-1}$ and $\Delta\widehat{R}_{i,j+1}^{e}$ has the same reputation and $\Delta\widehat{R}_{i,j}^{e-1}$ appears in $rt_{\Delta R}^{e-j}$. The verification process is similar to that of the basic one.

VI. Analysis

A. Privacy Analysis

In privacy-preserving credit-based PoW and reputation-based BFT, the leader is elected via the ZKP. Thus, it does not break the information-theoretic privacy of committed values.

In the PIU, the privacy of the newly committed value is also guaranteed by the ZKP.

For the privacy-preserving reputation, the privacy of commitments $\widehat{ID}_i^e$ and $\widehat{R}_{i,0}^e$ is guaranteed by the perfect hiding property of the Pedersen commitment, i.e., the adversary cannot distinguish between tuples of $(R_a, g, h, g^{R_a}h^{r_a})$ and $(R_b, g, h, g^{R_a}h^{r_a})$. The property holds if r^a is randomly chosen from a uniform distribution.

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

Considering the adversary $\mathscr{A}_{pr}$ can know ΔR_i^e of a validator i during consensus, we prove such information will not break the privacy of the reputation commitment by contradiction. Specifically, if $\mathscr{A}_{pr}$ can break privacy, i.e., it can distinguish the tuples of the form $(R_i^e, \Delta R_i^e, g, h, g^{R_i^e} h^{r_i^e}, g^{R_i^e + \Delta R_i^e} h^{r_i^e})$ and $(R_j^e, \Delta R_i^e, g, h, g^{R_i^e} h^{r_i^e}, g^{R_i^e + \Delta R_i^e} h^{r_i^e})$ for a randomly chosen r_i^e, then we can use $\mathscr{A}_{pr}$ to construct an adversary $\mathscr{A}_{ph}$ that breaks the perfect hiding property of the Pedersen commitment as follows: after receiving the challenge (x, g, h, y) where (x, y) is distributed either as $(R_1, g^{R_1} h^{r_1})$ or as $(R_2, g^{R_1} h^{r_1})$, $\mathscr{A}_{ph}$ prepares ΔR_1 and calls $\mathscr{A}_{pr}$ with input $(x, \Delta R_1, g, h, y, y \cdot g^{\Delta R_1})$. Then, $\mathscr{A}_{pr}$ returns the $k \in \{1, 2\}$. If k is correct, $\mathscr{A}_{ph}$ returns R_1 or else it returns R_2.

B. Security Analysis

zkRep is proved to defend against the following attacks.

1) Reputation Manipulation Attack: An attacker tries to modify the real reputation value in the reputation commitments to increase its chances to be elected as a leader.

Since the underlying blockchain guarantees that all the reputation commitments will achieve consensus, the attacker cannot propose a reputation commitment arbitrarily. Thus, we need to prove the integrity of R^i in zkRep when all the reputation commitments are recorded on the chain.

First, the newcomer, validator i, will join the zkRep with an initial score R_{init} at epoch $e-1$. Thus, its reputation commitment is $\mathsf{COMM}(R_{\text{init}} + q, r_i^{e-1})$ and the integrity of which is guaranteed by the π_{NEW}. The validator i works from the next epoch after the PIU protocol. Since π_{PIU} needs to prove that $R_{i,0}^e$ equals R_i^{e-1} and its reputation commitment is $\mathsf{COMM}(R_{i,0}^e + q, r_i^e)$, validator i cannot modify the score during epoch transition. Then within epoch e, validator i earns the reputation ΔR_i^e, which achieves the consensus. The reputation commitment of i can be recorded in the block as $\widehat{R}_{i,k}^e = \mathsf{COMM}(R_{i,0}^e + \sum_{j=0}^{k} \Delta R_{i,j}^e + q, r_i^e)$. It can be verified by other validators due to the additive homomorphic property of the Pedersen commitment. At the end of epoch e, validator i will change the ID and reputation commitment via the PIU protocol.

2) Sybil Attack: Such attackers may try to join the system by generating multiple identities illegitimately.

zkRep requires the new comer to join the system with an identity as well as a PoW puzzle's solution. Suppose the attacker wants to join in epoch e, it needs to generate multiple solutions of the PoW puzzle related to the global random seed. The global random seed can only be generated just before the beginning of epoch $e-1$. As a result, the attacker can hardly infer the seed before epoch $e-1$ and generate multiple identities within a short time.

3) Identity Fraud: The attacker steals the ID of victim i, i.e., get the rs_i to inherit its reputation after compromising it.

Once the victim has recovered and known attack events, it will publish the root secret as well as the previous commitments. Other validators will add the leaked root secret in the blacklist BL so that the attacker can never generate a proof π_{PIU} that has a root secret within the BL.

4) Side Channel Attack: Although the adversary cannot obtain the reputation of the victim from its reputation commitments, it can still breach the privacy through a side channel. For example, the time of mining a block is correlated with the validator's reputation in the credit-based PoW. In reputation-based BFT, winning the lottery depends on its reputation.

Therefore, with a longer epoch length (i.e., a larger number of blocks within an epoch), it is more likely for the adversary to find high reputation validators. We analyze the appropriate epoch length l via Bayes' theorem as follows.

First, we assume the reputation scores of all the validators follow some distribution in epoch e. Based on this, we can calculate the probability $P_1, P_2, \ldots, P_n$ that each validator wins the lottery or mines the block in the reputation-based blockchain. We define the set V^{H_e} as the set of high reputation validators, i.e., their reputation values are in the top $1/3$ of all the values of validators V^{n_e}. We define event A as the validator i is a high reputation validator. We define the observation O as the number of times u where a validator wins the lottery or mines a block.

Now, the adversary can calculate the probability $P(A|O)$ by using the Bayesian inference as follows:

$$P(A|O) = \frac{P(O|A) \cdot P(A)}{P(O|A) \cdot P(A) + P(O|\neg A) \cdot P(\neg A)} = \frac{\sum_{i \in V^{H_e}} P(O|i)}{\sum_{i \in V^{n_e}} P(O|i)}$$

where $P(O|i) = \sum_{j=u}^{l} \binom{l}{u} P_i^j (1 - P_i)^{l-j}$.

Then, we can define event B as the set of all victims V^{x_e} contains x high reputation nodes at the end of the epoch e. Since the adversary can only choose the victims based on O, the probability $P(B) = \prod_{i \in V^{x_e}} P(O|i)$.

The adversary wins if three conditions are satisfied: given thresholds T_1 and T_2, 1) $P(A) > T_1$; 2) $P(B) > T_2$; and 3) there is enough time for the adversary to compromise the targets after conditions 1) and 2) are satisfied. Since it is hard to quantify time in condition 3), we set appropriate epoch length l to be less than the time required for conditions 1) and 2).

We provide two cases with parameters in Section VII. The reputation distribution is assumed as a uniform distribution, $x = \lfloor 1/3n \rfloor, T_1 = 1/3, T_2 = 1/2$.

Case 1: Credit-based PoW, $n = 1000$: l should be smaller than the time of generating 2894 blocks. Thus, one epoch takes around two days, since the block generation interval is around 50 s (See Section VII-D).

Case 2: Reputation-based BFT, $n = 100$: l should be smaller than the time for 177 times of the leader election. Thus, one epoch takes time at the order of days to weeks, since a leader can work from minutes to hours in BFT protocol.

C. Integration With Existing Reputation-Based Blockchain

First, we need to consider the epoch manner. B-IoT, PoT, and RepuCoin are not epoch-based blockchains. To integrate with them, we need to add new blocks in their design. B-IoT is not a chain-structured blockchain but a DAG structured blockchain named IOTA Tangle [27]. We can leverage a special transaction named milestone. IOTA defines that a transaction is confirmed if and only if it is referenced by a milestone. Thus, every validator can see the milestones. We can define two special milestones as the beginning and end

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

of an epoch, respectively, and use them to store the necessary information in our identity update protocol, such as the Merkle-trees of ID and reputation commitments. For chain-based blockchains, such as RepuCoin and PoT, we can add a state block that represents the beginning and end of an epoch.

Then, we need to consider the integration with their consensuses within the epoch. B-IoT and PoT can directly utilize the privacy-preserving PoW (PPOW)in zkRep. RepChain is based on reputation-based BFT which can be modified to the privacy-preserving BFT in zkRep. However, RepuCoin uses reputation weighted BFT which treats the weight of each vote as its reputation over the total reputation of the group. The privacy-preserving weighted voting has been discussed and implemented in [28] and [29]. zkRep can enhance other parts of it, including the privacy-preserving reputation and the PIU protocol.

Finally, the epoch transition should be handled. B-IoT and RepuCoin do not need additional operations during epoch transition. Since PoT uses the trust links between validators to represent reputation, it adopts the PageRank algorithm to update the reputation. Similarly, RepChain needs a sorting algorithm to shard the validators during epoch transition. These algorithms are hard to be realized based on the Pedersen commitment. One possible way is to reveal the reputation and run the algorithm, i.e., PageRank or sorting, once per epoch. Since the time for identity update protocol especially the validation of the $\pi_{\mathtt{PIU}}$ is quite short according to our evaluation. Thus, a validator can generate the proof first and then reveal the reputation, making it infeasible for the adversary to attack in such a limited time. Another possible way is to leverage state-of-the-art homomorphic encryption [30], [31].

VII. Evaluation

A. Implementation and Evaluation Setup

We implement zkRep in C++ and Go. The $\pi_{\mathtt{PPOW}}$, $\pi_{\mathtt{PLE}}$, and $\pi_{\mathtt{PIU}}$ are implemented in C++ based on libsnark.[1] We implement a synchronous reputation-based BFT consensus [32] that is used in RepChain [4] in Go. For reputation-based PoW consensus, we simulate it in Go.

We evaluate zkRep in C5.Large instances on AWS. Each instance is equipped with 4-GB RAM and 2 vCPUs with second generation Intel Xeon Scalable Processors (3.6 GHz). The bandwidth of each instance is set as 20 Mb/s.

Since the measurement of reputation is not our focus, we simply calculate the reputation based on the votes or the blocks from validators. In BFT consensus, the validator that votes for the transaction in a valid block will earn one reputation score. While in PoW consensus, the validator who mines the block will earn one reputation score. For the reputation decay mechanism, we set $\eta = 0.5$ and $\alpha = 10$ for each epoch. We set the sliding window $w = 8$. Some validators can be lazy or have limited resources. Thus, we define the capability of a validator as the ratio of transactions it can vote overall transactions it receives, which is assumed to be uniformly in the range from 5% to 100%. In our evaluation, we test validators have (1) the same capability, and (2) different capabilities.

[1]https://github.com/scipr-lab/libsnark

TABLE III
Performance of $\pi_{\mathtt{PLE}}$ and $\pi_{\mathtt{PPOW}}$

Proof	$\pi_{\mathtt{PLE}}$	$\pi_{\mathtt{PPOW}}$
size of pk	14.9MB	10.5MB
size of vk	821B	559B
KeyGen Time	5.3s	3.7s
Prove Time	5.5s	3.4s
Verify Time	0.003s	0.003s

TABLE IV
Prove/Verify time of $\pi_{\mathtt{PIU}}$ with Different w and n

w \ n	10	20	50	100
1	25.3s	31.5s	38.7s	41.6s
4	42.2s	52.3s	61.6s	73.9s
8	65.4s	80.2s	93.5s	106.8s
12	90.7s	107.8s	126.4s	152.0s

We test the blockchain with and without zkRep under slowly adaptive attack. Since the reputation is a commitment in zkRep, the attacker cannot distinguish the validators and will degrade to a random attacker that randomly selects victims during epoch transition with an appropriate epoch length.

B. Proof Overhead for zkRep

We measure the overhead of three zk-SNARK proofs: $\pi_{\mathtt{PPOW}}$, $\pi_{\mathtt{PLE}}$, and $\pi_{\mathtt{PIU}}$ with different time windows w and different numbers of validators n. The performance is shown in Tables III and IV. Since pk and the circuit for the proofs will only be created in the one-time setup, the relatively large size of pk, and long setup time are acceptable. For the $\pi_{\mathtt{PPOW}}$, the time of Prove and Verify is 3.4 and 0.003 s, respectively. Comparing with the mining time of several minutes, such overhead is negligible. For the $\pi_{\mathtt{PLE}}$, the time of Prove and Verify is 5.5 and is 0.003 s, respectively. They are still acceptable since the consensus latency is at the orders of seconds to minutes, e.g., 60 s for RepChain, several mins for RepuCoin. For the $\pi_{\mathtt{PIU}}$, the time of Prove is relevant to the n. The time complexity of proving $\pi_{\mathtt{PIU}}$ is dominated by the Merkle-tree proof, which takes $O(w \log n)$ time. All these proofs are used during epoch transitions that last a few hours [4], [7]. It is worth mentioning that the zk-SNAKR guarantees the constant proof size, which is 312 bytes for all proofs. Thus, such overhead can be ignored.

C. Performance Evaluation of Reputation-Based BFT

In this evaluation, we run 20 epochs of the reputation-based BFT-based blockchain with and without zkRep. All experiments are tested ten times with different $n = 10, 50, 100$ and two settings of validator's capability. We run n instances in AWS and record the throughput. We also snapshot the reputation distribution of validators in epoch 1, 5, 10, and 20 under different capability settings.

Fig. 3 shows the throughput in processed transaction per second (TPS), which is the one result among ten times experiments. For the parameter of Fig. 3(a), the average TPSs are 816.1 and 820.6 for zkRep and original blockchain, respectively, which means the overhead of proving $\pi_{\mathtt{PLE}}$ is large than

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

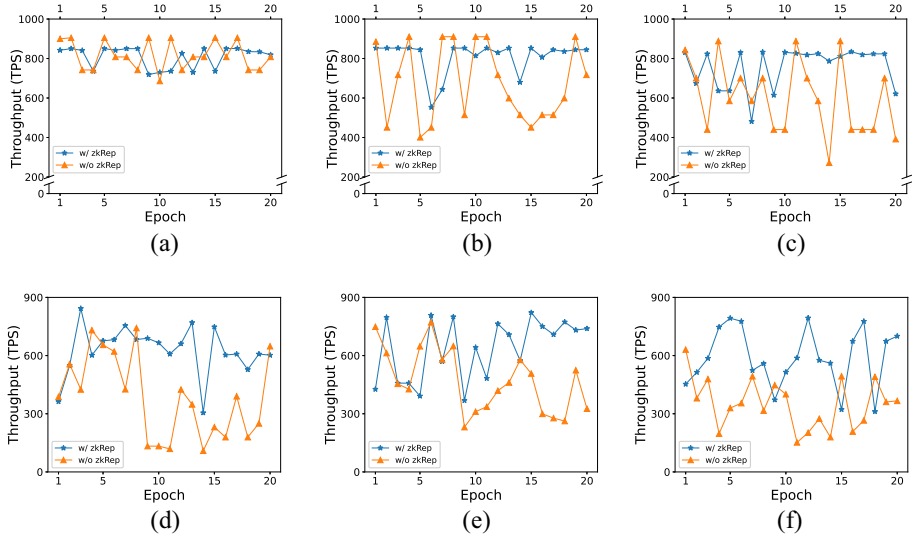

Fig. 3. Throughput of the reputation-based BFT consensus with and without zkRep under slowly adaptive attack. The capability of validators is the same in (a)–(c) but different in (d)–(f). (a) $n = 10$. (b) $n = 50$. (c) $n = 100$. (d) $n = 10$. (e) $n = 50$. (f) $n = 100$.

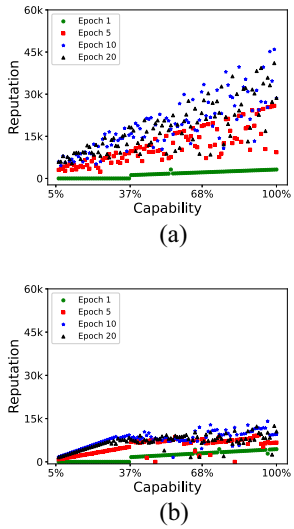

Fig. 4. Reputation distribution of 100 validators with different capabilities. (a) w/ zkRep. (b) w/o zkRep.

the cost of the leader election. However, with n increases, the situation changes. The average TPSs with and without zkRep are 811.2 and 655.3 with the same parameter as Fig. 3(b), 751.4 and 599.5 with the same parameter as Fig. 3(c). The performance gain mainly comes from preserving the privacy of reputation. In the original BFT protocol, the attacker has at least $1/3/1/2 = 2/3$ probability to control the leader and cause the leader election while this probability drops to $1/3$ in zkRep. The cost of the leader election will be much larger than the overhead of the π_{PLE} with large n. Considering different capabilities of validators, the situation will be much worse. The average TPSs with and without zkRep are 625.3 and 371.9, 637.0 and 470.2, and 590.7 and 375.0 with the same parameter as Fig. 3(d) and (f), respectively.

Fig. 4 shows the reputation distribution of the validators. It indicates that the reputation can still represent the capability with zkRep, i.e., the high capability validators have a high reputation. However, in Fig. 4(b), the high reputation validators will become the victims periodically so that their reputation will not exceed a threshold. Thus, the reputation can no longer represent the behavior and the capability of the validator.

D. Performance Evaluation of Credit-Based PoW

We test the credit-based PoW for 40 epochs with and without zkRep under slowly adaptive attack. We set 20 trust divisions: from TD_1 to TD_{20} and the reputation range for TD_i is $[i, i+1)$. As the mining difficulty inversely proportional to reputation and the latency for a block is 10 min in Bitcoin, validators in TD_i are set to mine the block with the latency of t_b/i and $t_b = 10$ min. We include the latency for Prove and Verify of π_{PPOW} which is 3.4 s in the simulation with zkRep. Recall that the highest f validators' reputation should be less than half the total reputation. Since we assume the uniform distribution of different capabilities, we set $f < 0.3n$.

In order to simulate the situation of low-difficulty PoW, we evaluate the credit-based PoW based on the Bernoulli trail similar to PoT [6]. First, we select a trust division i with the probability equal to $i/210$, i.e., the trust i represents over total trust. Then we randomly select a nonadversary validator from the highest TD that passed the Bernoulli trail. We also test it in the case of validators with different capabilities. The validator mines the block with the probability equal to its capability.

Fig. 5 shows the average latency for each epoch. Similar to the BFT-based blockchain, the latency of a block will increase without zkRep and the advantage will be amplified with n increases. The work [6] shows that most blocks are generated by the validators with a top 25% reputation. Thus, the latency of block generation is mostly determined by these validators. When a slowly adaptive attacker compromises validators with top 33% reputation, the tasks of block generation would be accomplished by validators within around 34% to 60% reputation range, which results in larger latency.

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

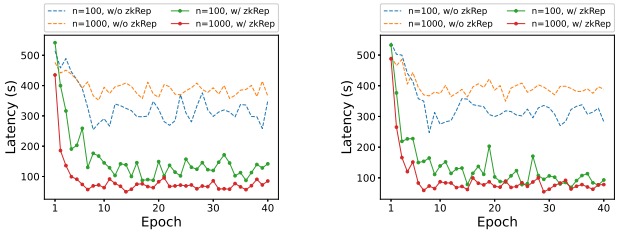

Fig. 5. Latency of the credit-based PoW consensus with and without zkRep under slowly adaptive attack. (a) Same capability. (b) Different capability.

TABLE V
EPOCH TRANSITION TIME WITH DIFFERENT w AND n

w \ n	10	20	50	100
1	46s	56s	67s	73s
4	65s	86s	97s	109s
8	88s	110s	140s	145s
12	113s	142s	173s	190s

E. Epoch Transition Time

We test the epoch transition time for identity update protocol with different w and n instances in AWS. The results are shown in Table V. The time will increase with larger n and w but no larger than 4 min. It indicates that our protocol will not introduce too much overhead to the system.

VIII. Future Work

A. Different Reputation Metrics

Reputation can be quantified by different metrics, such as the trust links between validators in PoT, or the overall feedback from other validators [33], which lead to different operations in reputation updating. For example, the feedback type requires the multiplication and addition operations on different validators' reputation commitments. One possible way is to leverage the state-of-the-art homomorphic encryption [30], [31].

B. Anonymous Communication

The attacker may track the IP address to link the validators across the epoch. To have a fully anonymous property, we should let all the validators run over the anonymous communication layer such as Onion Routing (Tor) supported by existing works [34], [35].

C. Different Zero-Knowledge Proof Protocols

zkRep adopts the widely-used ZKP protocol, zk-SNARKs for proof-of-concept, which is not perfect in terms of the need for a trusted setup and inefficiency on some operations. Other protocols, such as bulletproof [36], zkLedger [17] can be considered to overcome these issues in future work.

IX. Conclusion

In this article, we propose a privacy-preserving scheme, zkRep, to protect the reputation-based blockchains. zkRep leverages the Pedersen commitment-based reputation updating protocol and two PLE approaches to operation on committed reputation. Moreover, the ZKP-based identity update protocol can change the ID and reputation commitments to defend against the slowly adaptive attack. The analysis and experimental results demonstrate that zkRep effectively prevents the slowly adaptive attacks without inducing too much overhead.

Acknowledgment

Any opinions, findings and conclusions or recommendations expressed in this material are those of the authors and do not reflect the views of National Research Foundation, Singapore.

References

[1] O. Novo, "Blockchain meets IoT: An architecture for scalable access management in IoT," *IEEE Internet Things J.*, vol. 5, no. 2, pp. 1184–1195, Apr. 2018.

[2] Z. Yang, K. Yang, L. Lei, K. Zheng, and V. C. M. Leung, "Blockchain-based decentralized trust management in vehicular networks," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 1495–1505, Apr. 2019.

[3] J. Huang, L. Kong, G. Chen, L. Cheng, K. Wu, and X. Liu, "B-IoT: Blockchain driven Internet of Things with credit-based consensus mechanism," in *Proc. ICDCS*, 2019, pp. 1348–1357. [Online]. Available: https://doi.org/10.1109/ICDCS.2019.00135

[4] C. Huang *et al.*, "RepChain: A reputation based secure, fast and high incentive blockchain system via sharding," *IEEE Internet Things J.*, vol. 8, no. 6, pp. 4291–4304, Mar. 2021.

[5] J. Yu, D. Kozhaya, J. Decouchant, and P. Verissimo, "RepuCoin: Your reputation is your power," *IEEE Trans. Comput.*, vol. 68, no. 8, pp. 1225–1237, Aug. 2019.

[6] L. Bahri and S. Girdzijauskas, "Trust mends blockchains: Living up to expectations," in *Proc. ICDCS*, 2019, pp. 1358–1368. [Online]. Available: https://doi.org/10.1109/ICDCS.2019.00136

[7] M. Zamani, M. Movahedi, and M. Raykova, "RapidChain: Scaling blockchain via full sharding," in *Proc. CCS*, 2018, pp. 931–948. [Online]. Available: https://doi.org/10.1145/3243734.3243853

[8] E. Kokoris-Kogias, P. Jovanovic, L. Gasser, N. Gailly, E. Syta, and B. Ford, "OmniLedger: A secure, scale-out, decentralized ledger via sharding," in *Proc. IEEE SP*, 2018, pp. 583–598. [Online]. Available: https://doi.org/10.1109/SP.2018.000-5

[9] E. Kokoris-Kogias, P. Jovanovic, N. Gailly, I. Khoffi, L. Gasser, and B. Ford, "Enhancing bitcoin security and performance with strong consistency via collective signing," in *Proc. USENIX Security Symp.*, 2016, pp. 279–296. [Online]. Available: https://www.usenix.org/conference/usenixsecurity16/technical-sessions/presentation/kogias

[10] A. Kiayias, A. Russell, B. David, and R. Oliynykov, "Ouroboros: A provably secure proof-of-stake blockchain protocol," in *Proc. CRYPTO*, 2017, pp. 357–388. [Online]. Available: https://doi.org/10.1007/978-3-319-63688-7_12

[11] G. Maxwell. (2013). *CoinJoin: Bitcoin Privacy for the Real World*. [Online]. Available: https://bitcointalk.org/index.php?topic=279249.0

[12] T. Ruffing, P. Moreno-Sanchez, and A. Kate, "CoinShuffle: Practical decentralized coin mixing for bitcoin," in *Proc. ESORICS*, 2014, pp. 345–364. [Online]. Available: https://doi.org/10.1007/978-3-319-11212-1_20

[13] N. Van Saberhagen. (2013). *Cryptonote V 2.0*. [Online]. Available: https://bytecoin.org/old/whitepaper.pdf

[14] S. Sun, M. H. Au, J. K. Liu, and T. H. Yuen, "RingCT 2.0: A compact accumulator-based (linkable ring signature) protocol for blockchain cryptocurrency monero," in *Proc. ESORICS*, 2017, pp. 456–474. [Online]. Available: https://doi.org/10.1007/978-3-319-66399-9_25

[15] I. Miers, C. Garman, M. Green, and A. D. Rubin, "Zerocoin: Anonymous distributed e-cash from bitcoin," in *Proc. IEEE SP*, 2013, pp. 397–411. [Online]. Available: https://doi.org/10.1109/SP.2013.34

[16] E. Ben-Sasson *et al.*, "Zerocash: Decentralized anonymous payments from bitcoin," in *Proc. IEEE SP*, 2014, pp. 459–474. [Online]. Available: https://doi.org/10.1109/SP.2014.36

[17] N. Narula, W. Vasquez, and M. Virza, "zkLedger: Privacy-preserving auditing for distributed ledgers," in *Proc. NSDI*, 2018, pp. 1–16. [Online]. Available: https://www.usenix.org/conference/nsdi18/presentation/narula

[18] T. Kerber, A. Kiayias, M. Kohlweiss, and V. Zikas, "Ouroboros crypsinous: Privacy-preserving proof-of-stake," in *Proc. IEEE SP*, 2019, pp. 157–174. [Online]. Available: https://doi.org/10.1109/SP.2019.00063

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

[19] C. Ganesh, C. Orlandi, and D. Tschudi, "Proof-of-stake protocols for privacy-aware blockchains," in *Advances in Cryptology - EUROCRYPT2019 - 38th Annual International Conference on the Theory and Applications of Cryptographic Techniques*, Darmstadt, Germany, May, 2019, [Online]. Available: https://doi.org/10.1007/978-3-030-17653-223

[20] F. Baldimtsi, V. Madathil, A. Scafuro, and L. Zhou, "Anonymous lottery in the proof-of-stake setting," in *Proc. IEEE 33rd Comput. Security Found. Symp. (CSF)*, 2020, pp. 318–333.

[21] S. Nakamoto. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*. [Online]. Available: https://bitcoin.org/bitcoin.pdf.

[22] Y. Gilad, R. Hemo, S. Micali, G. Vlachos, and N. Zeldovich, "Algorand: Scaling Byzantine agreements for cryptocurrencies," in *Proc. SOSP*, 2017, pp. 51–68. [Online]. Available: https://doi.org/10.1145/3132747.3132757

[23] N. Bitansky, A. Chiesa, Y. Ishai, R. Ostrovsky, and O. Paneth, "Succinct non-interactive arguments via linear interactive proofs," in *Theory of Cryptography - 10th Theory of Cryptography Conference (TCC)*,Tokyo, Japan, Mar. 2013. [Online]. Available: https://doi.org/10.1007/978-3-642-36594-218

[24] V. Nikolaenko, U. Weinsberg, S. Ioannidis, M. Joye, D. Boneh, and N. Taft, "Privacy-preserving ridge regression on hundreds of millions of records," in *Proc. IEEE SP*, 2013, pp. 334–348. [Online]. Available: https://doi.org/10.1109/SP.2013.30

[25] Q. Wang *et al.*, "Privacy-preserving collaborative model learning: The case of word vector training," *IEEE Trans. Knowl. Data Eng.*, vol. 30, no. 12, pp. 2381–2393, Dec. 2018. [Online]. Available: https://doi.org/10.1109/TKDE.2018.2819673

[26] E. Syta *et al.*, "Scalable bias-resistant distributed randomness," in *Proc. IEEE Symp. Security Privacy (SP)*, 2017, pp. 444–460.

[27] S. Popov, "The tangle," Berlin, Germany, IOTA Found., White Paper, 2016.

[28] B. Bünz, S. Agrawal, M. Zamani, and D. Boneh, "Zether: Towards privacy in a smart contract world," in *Proc. Int. Conf. Financ. Cryptogr. Data Security*, 2020, pp. 423–443.

[29] P. McCorry, S. F. Shahandashti, and F. Hao, "A smart contract for boardroom voting with maximum voter privacy," in *Proc. FC*, 2017, pp. 357–375. [Online]. Available: https://doi.org/10.1007/978-3-319-70972-7_20

[30] D. Catalano and D. Fiore, "Using linearly-homomorphic encryption to evaluate degree-2 functions on encrypted data," in *Proc. CCS*, 2015, pp. 1518–1529. [Online]. Available: https://doi.org/10.1145/2810103.2813624

[31] P. Martins, L. Sousa, and A. Mariano, "A survey on fully homomorphic encryption: An engineering perspective," *ACM Comput. Surveys*, vol. 50, no. 6, pp. 1–33, 2017.

[32] L. Ren, K. Nayak, I. Abraham, and S. Devadas, "Practical synchronous Byzantine consensus," 2017. [Online]. Available: arXiv:1704.02397.

[33] L. Xiong and L. Liu, "PeerTrust: Supporting reputation-based trust for peer-to-peer electronic communities," *IEEE Trans. Knowl. Data Eng.*, vol. 16, no. 7, pp. 843–857, Jul. 2004. [Online]. Available: https://doi.org/10.1109/TKDE.2004.1318566

[34] (2020). *Tor Support in Zcash*. [Online]. Available: https://zcash.readthedocs.io/en/latest/rtd_pages/tor.html

[35] (2020). *Monero—Github*. [Online]. Available: https://github.com/monero-project/monero#using-tor

[36] B. Bünz, J. Bootle, D. Boneh, A. Poelstra, P. Wuille, and G. Maxwell, "Bulletproofs: Short proofs for confidential transactions and more," in *Proc. IEEE Symp. Security Privacy (SP)*, 2018, pp. 315–334.

Chenyu Huang received the B.S. degree in computer science from Wuhan University, Wuhan, China, in 2015, and the Ph.D. degree from Hong Kong University of Science and Technology, Hong Kong, in 2020.

He is currently a Researcher with Ping An Technology, Shenzhen, China. His research interests include IoT security, blockchain, and federated learning.

Yongjun Zhao (Member, IEEE) received the B.Eng. (with First-Class Hons.), in 2019, and Ph.D. degrees from the Department of Information Engineering, Chinese University of Hong Kong, Hong Kong, in 2018.

He is currently doing a Postdoctoral Research with The Strategic Centre for Research in Privacy-Preserving Technologies and Systems, Nanyang Technological University, Singapore. He has published in AAAI, NDSS, PKC, ESORICS, Financial Crypt, PoPETs, and AsiaCCS. His main research interests are applied cryptography, cryptographic cloud storage, privacy-enhancing technologies, and blockchain.

Dr. Zhao served on the program committees of many conferences, including *PoPETs*, CANS, InsCrypt, *Neutrosophic Sets and Systems*, SECRYPT, SCC, and SecureComm.

Huangxun Chen received the the B.S. degree in computer science from Shanghai Jiao Tong University, Shanghai, China, in 2015, and the Ph.D. degree in computer science from Hong Kong University of Science and Technology, Hong Kong, in 2020.

She is currently a Researcher with the Theory Lab of 2012 Labs, Huawei, Shenzhen, China.

Xu Wang received the B.S. degree in information security and the master's degree in computer science and technology from Wuhan University, Wuhan, China, in 2018 and 2021, respectively.

His research interests include blockchain technology and game theory.

Qian Zhang (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees in computer science from Wuhan University, Wuhan, China, in 1994, 1996, and 1999, respectively.

She joined Hong Kong University of Science and Technology (HKUST), Hong Kong, in September 2005 where she is currently a Tencent Professor of Engineering and the Chair Professor with the Department of Computer Science and Engineering. She is also serving as the Co-Director of Huawei-HKUST Innovation Lab and the Director of Digital Life Research Center, HKUST. Before that, she was in Microsoft Research Asia, Beijing, China, from July 1999, where she was the Research Manager of the Wireless and Networking Group. She has published more than 400 refereed papers in international leading journals and key conferences. Her current research interests include Internet of Things, smart health, mobile computing and sensing, wireless networking, and cyber security.

Dr. Zhang is serving as the Editor-in-Chief of IEEE TRANSACTIONS ON MOBILE COMPUTING. She is a member of Steering Committee of IEEE Infocom.

Yanjiao Chen (Senior Member, IEEE) received the B.E. degree in electronic engineering from Tsinghua University, Beijing, China, in 2010, and the Ph.D. degree in computer science and engineering from Hong Kong University of Science and Technology, Hong Kong, in 2015.

She is currently a Bairen Researcher with Zhejiang University, Hangzhou, China. Her research interests include network security, AI security, and wireless spectrum management.

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.

Huaxiong Wang received the Ph.D. degree in mathematics from the University of Haifa, Haifa, Israel, in 1996, and the Ph.D. degree in computer science from the University of Wollongong, Wollongong, NSW, Australia, in 2001.

He is currently an Associate Professor with the Division of Mathematical Sciences, School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore. His research interests include cryptography, information security, coding theory, combinatorics, and theoretical computer science.

Kwok-Yan Lam (Senior Member, IEEE) received the B.Sc. degree (first Class Hons.) from the University of London, London, U.K., in 1987, and the Ph.D. degree from the University of Cambridge, Cambridge, U.K., in 1990.

He is currently a Professor with the School of Computer Science and Engineering, Nanyang Technological University (NTU), Singapore, and the Director of the Strategic Centre for Research in Privacy-Preserving Technologies and Systems. From August 2020, he is also on part-time secondment to the INTERPOL as a Consultant of Cyber and New Technology Innovation. Prior to joining NTU, he has been a Professor with Tsinghua University, Beijing, China, from 2002 to 2010 and a Faculty Member of the National University of Singapore, Singapore, and the University of London, since 1990. He was a Visiting Scientist with the Isaac Newton Institute, Cambridge University, Cambridge, U.K., and a Visiting Professor with the European Institute for Systems Security. His research interests include distributed systems, intelligent systems, IoT security, distributed protocols for blockchain, homeland security, and cybersecurity.

Prof. Lam received the Singapore Foundation Award in 1998 from the Japanese Chamber of Commerce and Industry in recognition of his research and development achievement in information security in Singapore.

Authorized licensed use limited to: Nanyang Technological University. Downloaded on September 15,2022 at 18:48:40 UTC from IEEE Xplore. Restrictions apply.